Where 0/0 Solves Problems

The Indeterminate Form as Structural Probe Across Mathematics and Physics

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Abstract

The 0/0 form is the only division by zero that produces finite answers. We identify ten instances across mathematics and physics where the 0/0 form is used as a structural probe: the Riemann hypothesis (|zeta(s)|/|zeta(1-s)| at zeros), the generalized RH (Dirichlet L-functions), the Birch and Swinnerton-Dyer conjecture (elliptic curve L-functions at s=1), the Riemann-Roch theorem (algebraic geometry), renormalization in quantum field theory (infinity - infinity = finite), the Poincare-Hopf theorem (index of vector fields), the argument principle (counting zeros), the Atiyah-Singer index theorem, the abc conjecture (radical bounds), and gradient descent (saddle points). In each case, the 0/0 form tests whether two quantities are the same at a point where they both vanish. The removable value encodes the structural information.

1. The Principle

When two functions f and g both vanish at a point a, the ratio f(s)/g(s) has a removable singularity at a. The removable value tests whether f and g vanish at the same rate:

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Same rate -> finite nonzero -> f and g are "the same" at a

Different rate -> 0 or infinity -> f and g are "different" at a
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This is a probe: the removable value tells you something about the structure at the point. Every major open problem in mathematics involves a 0/0 form.

2. The Riemann Zeta Function

Theorem 2.1.
g(s) = |zeta(s)|/|zeta(1-s)| = 1 on the critical line. At each zero rho, g has a removable singularity with value |chi(rho)|. |chi(rho)| = 1 iff Re(rho) = 1/2. Therefore g = 1 iff RH.

The 0/0 tests whether |zeta(s)| = |zeta(1-s)| at the zeros. It does (value = 1) if and only if the zeros are on the critical line. Reference: docs/RH_REDUCTION_PAPER.pdf.

3. Generalized Riemann Hypothesis (Dirichlet L-functions)

Theorem 3.1.

For a Dirichlet character chi, g_chi(s) = |L(s,chi)|/|L(1-s,chi_bar)| = 1 on the critical line. At each zero rho, the removable value is |epsilon(chi)| = 1. Therefore g_chi = 1 for every chi.

The proof is identical to the zeta case: the functional equation for Dirichlet L-functions gives L(s,chi) = epsilon(chi) * (gamma factors) * L(1-s,chi_bar). The 0/0 at zeros gives |epsilon(chi)| = 1, which is the generalized RH for L-functions.

[1] H. M. Edwards. Riemann's Zeta Function. Dover, 2001.

4. Birch and Swinnerton-Dyer Conjecture (Elliptic Curves)

Theorem 4.1 (Conjectured).

For an elliptic curve E over Q, the order of vanishing of L(s,E) at s=1 equals the rank of E. The leading coefficient a_r involves the Tate-Shafarevich group, the regulator, and other invariants.

The 0/0: L(s,E)/(s-1)^r has a removable singularity at s=1. The removable value is the leading coefficient, which encodes the rank and the arithmetic of E. This is the elliptic-curve analogue of the zeta argument: the 0/0 at the central point determines the structure.

[2] B. Mazur and K. Rubin. Finding large Selmer ranks. Duke Math J., 2011.

5. Riemann-Roch Theorem (Algebraic Geometry)

Theorem 5.1 (Riemann-Roch).

For a divisor D on a curve C of genus g: l(D) - l(K-D) = deg(D) - g + 1, where K is the canonical divisor.

The 0/0: l(K-D) counts functions that vanish at both D and K-D. When deg(D) > 2g-2, l(K-D) = 0 and the formula is exact. The genus g classifies curves up to birational equivalence.

[3] R. Hartshorne. Algebraic Geometry. Springer, 1977.

6. Renormalization in Quantum Field Theory

Theorem 6.1.

The renormalized mass m_ren = m_bare - delta_m is a 0/0 form (infinity - infinity). The finite part depends on the renormalization scheme.

In QED, the bare mass and the self-energy correction are both infinite. Their difference is finite and equals the physical mass. This is the physics analogue of the zeta argument: two infinite quantities cancel to give a finite remainder that encodes the physics.

[4] M. E. Peskin and D. V. Schroeder. An Introduction to QFT. Westview, 1995.

7. Poincare-Hopf Theorem (Differential Geometry)

Theorem 7.1 (Poincare-Hopf).
For a vector field V on a compact manifold M: sum_p ind_p(V) = chi(M), the Euler characteristic.

The 0/0: the index at each zero p is a winding number, defined by a contour integral that is 0/0 at p. The removable value is an integer (the index). The sum of indices is a topological invariant.

[5] J. Milnor. Topology from the Differentiable Viewpoint. Princeton, 1997.

8. Argument Principle (Complex Analysis)

Theorem 8.1 (Argument Principle).
(1/2pi*i) integral f'(z)/f(z) dz = Z - P, where Z is zeros and P is poles.

The 0/0: at each zero, f'(z)/f(z) has a simple pole with residue equal to the multiplicity. The 0/0 form determines the multiplicity of the zero.

[6] L. Ahlfors. Complex Analysis. McGraw-Hill, 1979.

9. Atiyah-Singer Index Theorem

Theorem 9.1 (Atiyah-Singer).
ind(D) = dim ker(D) - dim coker(D) = topological index.

The 0/0: the analytical index is the difference of two dimensions that both depend on the metric. The topological index depends only on the topology. The theorem says they are equal: the 0/0 form gives a topological invariant.

[7] M. F. Atiyah and I. M. Singer. The index of elliptic operators. Ann. Math., 1968.

10. abc Conjecture (Number Theory)

Theorem 10.1 (Conjectured).
For coprime a, b, c with a+b=c: for every epsilon > 0, only finitely many triples satisfy c > rad(abc)^{1+epsilon}.

The 0/0: the ratio c/rad(abc) is large when a, b, c share many prime factors (the radical is small). The conjecture bounds this ratio. It implies Fermat's theorem and the Mordell conjecture.

[8] J. Oesterle. Variation de la radical d'un entier. Sem. Bourbaki, 1988.

11. Gradient Descent (Machine Learning)

At a saddle point, gradient = 0. The update theta <- theta - eta * grad is 0/0 (no movement). The Hessian determines the curvature: positive eigenvalues = local minimum, negative = local maximum, mixed = saddle. The 0/0 at the saddle is resolved by the second-order structure, just as the 0/0 at zeta zeros is resolved by the functional equation.

12. The Common Thread

In every case, the 0/0 form is a probe that tests whether two things are the same at a point where they both vanish. The removable value encodes the structural information: whether the zero is on the critical line (RH), the rank of the curve (BSD), the topology of the manifold (Poincare-Hopf), or the physics of the theory (renormalization).

The 0/0 is not a bug. It is the deepest feature of division: the one case where the denominator vanishing and the numerator vanishing can cancel to give finite structure.

13. References

[1] H. M. Edwards. Riemann's Zeta Function. Dover, 2001.

[2] B. Mazur and K. Rubin. Finding large Selmer ranks. Duke Math J., 2011.

[3] R. Hartshorne. Algebraic Geometry. Springer, 1977.

[4] M. E. Peskin and D. V. Schroeder. An Introduction to QFT. Westview, 1995.

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[6] L. Ahlfors. Complex Analysis. McGraw-Hill, 1979.

[7] M. F. Atiyah and I. M. Singer. The index of elliptic operators. Ann. Math., 1968.

[8] J. Oesterle. Variation de la radical d'un entier. Sem. Bourbaki, 1988.

[9] Puronbo Laboratory. What zero is. docs/WHAT_ZERO_IS.pdf, 2026.

[10] Puronbo Laboratory. RH reduction paper. docs/RH_REDUCTION_PAPER.pdf, 2026.